A-6. Drainage Improvement

Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures
Agriculture Land Conservation	Medium	High		L Term

Descriptions

Measures	Improvement of drainage condition in the hilly terrain is crucial for the agricultural practices in the region as it contributes to maintaining soil health, preventing erosion, and ensuring optimum water content in the soil for crop growth. Effective drainage systems help manage excess water from rainfall, reduce the risk of waterlogging, and maintain soil structure, which is essential for sustainable agriculture. Contour plowing and terracing, installing subsurface drainage systems, creating drainage ditches and swales are some of the techniques which can improve drainage condition in the hilly agriculture lands of CHTs. Plowing along the contour lines of the slope creates natural barriers that slow down water runoff, allowing water to seep into the soil rather than flowing downhill and prevent erosion. Terracing creates flat areas on steep slopes where water can be more easily absorbed thus prevent erosion. Installing underground pipes or drainage tiles at strategic points in the field allows excess water to be channeled away from crop roots and reduces the risk of waterlogging. Similarly, shallow ditches or swales are created along the lower sections of the field or around the perimeter to direct water away from the crop area. These techniques support agricultural production while conserving soil and hill terrain.

Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
	■	■	■	■	■	■	■	■	■

Applicable Sites	Suitable for degraded lands, hill slopes

Slope	Less than 3%	3% to 15%	15% to 30%	30% to 60%	Above 60%
	√	√			

Design options and performance	• Gravity-Fed Contour Canals, Suitable for Slopes between 2-8%. • Lined Canals (Concrete, HDPE, or Stone Masonry), suitable for Steep slopes (above 5%) to reduce seepage and erosion. • Step-Down Canal with Drop Structures suitable for Slopes above 8%. • Terraced Irrigation Systems suitable for Sloping agricultural land.
Feasibility criteria	**Technical Feasibility** • Ensures slope stability, proper soil conditions, and adequate water availability for efficient irrigation without excessive erosion or seepage.

	Environmental Feasibility • Prevents erosion and land degradation through sustainable water flow management. • Maintains ecological balance by minimizing habitat disruption. • Reduces wastage and promotes efficient water use in hilly terrains. **Economic Feasibility** • Balances affordability and durability (e.g., earthen vs. lined canals). • Increases agricultural productivity, improving farmers' incomes. • Uses locally available materials and community-based management to reduce upkeep costs. **Social Feasibility** • Engages local farmers in planning, implementation, and maintenance. • Enhances food security and economic opportunities. • Aligns with government policies and local governance structures for long-term sustainability.
Operation and maintenance	• **Routine Maintenance** – Desilting, vegetation control, erosion repair. • **Structural Repairs** – Crack sealing, check dam rehabilitation, slope stabilization. • **Seasonal Adjustments** – Monsoon embankment reinforcement, dry-season storage optimization. • **Community Involvement** – Water user groups, local government support, training programs.
Cost	Medium
Benefits	

Type	**Low**	**Medium**	**High**
Canopy Cover Increase			
Carbon Sequestration			
Crop Production Increase			
Crop Failure Risk Reduction			
Ensuring Food Security			
Fruit Production Increase			
Forestry Goods Production Increase			
Livestock Production Increase			
Fisheries Production Increase			
Alternative Livelihood Generation			
Increase of Income			
Soil Erosion Control			
Improved Soil Health			
Soil Fertility Increase			
Sustainable Water Supply			
Water Quality Improvement			
Water Purification			

	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	**Low**	**Medium**	**High**
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	• Contour plowing and terracing create natural barriers on slopes reducing runoff and allowing soil water retention which prevents soil erosion and landslides. • Subsurface drainage system helps maintain optimum soil moisture level which supports the agriculture production. • Shallow ditches or swales created along the lower sections of the field helps direct excess surface water, preventing waterlogging and reducing the need for costly irrigation systems. • Drainage of excess water from the land improves overall soil health and reduces the amount of runoff, leading to better drainage and healthier crops.			
Remarks	This measure will aid landscape restoration in CHT of Bangladesh along with benefitting the agriculture production.			